SysMon: System Monitor for Private Infrastructure

Requirements and System Design Specification

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1 Project Overview

Project Title: SysMon
Target Audience: Homelab enthusiasts and small-sized businesses wanting to utilize their own infrastructure.
Goal: To deliver a seamless user experience similar to Web UIs from enterprise solutions (Azure/AWS), allowing both technical and non-technical users to monitor, maintain, and build on private infrastructure.

1.1 Core Modules

1. **Frontend:** User Interface for visualization, configuration, and monitoring.
2. **Backend:** Orchestration of tasks, container management, telemetry aggregation, and host-level filesystem management.
3. **Authentication System:** Tier-based access control (Tier 1 to Tier 3) for secure team collaboration.

2 Key Features

- **Container Orchestration:** Support for both Docker and Podman. Features manual container creation via Compose files, alongside a visual UI builder that auto-generates Compose files.
- **Resource Governance (Resource Groups):** Tier 1 admins can create profiles limiting compute resources (e.g., thread limits, RAM caps, GPU access restrictions) enforced at the cgroup level via Compose deployments.
- **Storage Quotas (Storage Groups):** Provisioning of bounded host directories for container bind-mounts. Ensures specific user groups cannot exceed predetermined disk space limits.
- **Network Management:** Creating profiles, static/dynamic IP allocation, and a firewall rules manager.
- **Telemetry & Monitoring:** Real-time visualization of Memory utilization, CPU utilization, and Disk space.

3 User Stories

- **US1 (Resource Governance):** As a Tier 1 admin, I want to create a "Basic Resource Group" that restricts deployments to 2 CPU cores and 4GB of RAM, so that lower-tier users cannot accidentally exhaust the host server's resources.
- **US2 (Storage Quotas):** As a Tier 1 admin, I want to create a Storage Group with a 50GB quota, so that users deploying applications via the visual builder can only map volumes to this restricted folder without filling up the host system's root drive.

- **US3 (Visual Deployment):** As a Tier 3 (newer) admin, I want to visually attach a Docker image and select a pre-approved Resource Group and Storage Group, so that the system automatically generates a compliant Compose file.
- **US4 (Advanced Deployment):** As a Tier 1 admin, I want to paste my existing custom `docker-compose.yml` files directly into the system to easily migrate existing complex homelab stacks.

4 Crucial Use Case Description (Alistair Cockburn's Schema)

Use Case Name	**Visual Configuration and Deployment with Resource Limits**
Primary Actor	Admin (Tier 2 or Tier 3)
Scope	SysMon Orchestration & Frontend Module
Level	User Goal
Brief Description	The admin configures a new container via the UI, assigning it to a specific Resource Group and Storage Group. SysMon generates a Compose file injecting these limits and deploys it.
Preconditions	The admin is authenticated. Resource Groups and Storage Groups have been previously defined by a Tier 1 admin.
Success Guarantee	A Compose file is generated containing `deploy.resources` limits, volume paths are validated against the Storage Group, and the container is running.
Main Success Scenario	1. Admin navigates to "Create Container" and selects "Visual Builder". 2. Admin inputs the image repository tag. 3. Admin selects an assigned Resource Group 4. Admin maps persistent data to an assigned Storage Group folder. 5. Admin selects the target engine (Docker or Podman) and clicks Deploy. 6. SysMon Backend injects cgroup limits and approved host paths into a generated `docker-compose.yml` string. 7. SysMon executes the deployment via the API. 8. System begins streaming telemetry for the new container.
Extensions (Alternative Flows)	6a. Storage Group quota exceeded: System halts generation and notifies the admin that the selected storage pool is full. 7a. Engine daemon unreachable: System saves the generated Compose file as a draft and notifies the admin.

Table 1: Use Case: Visual Configuration with Governance

5 Class Diagram Specification

The following diagram presents the conceptual class structures focusing on visual builder configurations, Compose deployments, and resource/storage governance.

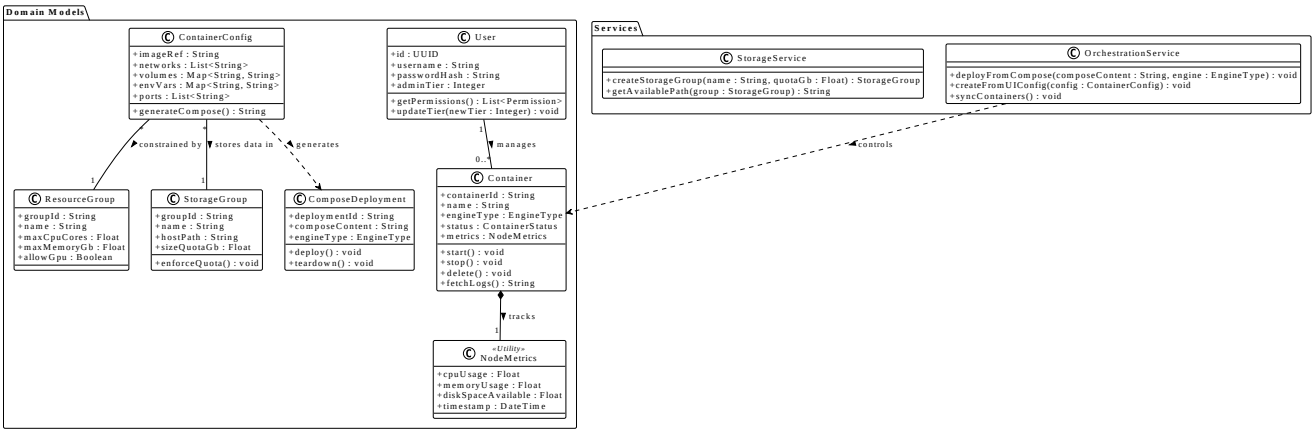


Figure 1: SysMon Domain and Service Class Diagram